



PROGRAM NAME: B.Tech CSE
YEAR/SEMESTER 4/7
FIRST CONTINUOUS ASSESSMENT, 2025-26

Course Name: Distributed Systems
Time : 1.30 hrs

Course Code: CS 410
Max. Marks : 20

- Note :** (i) The question paper contains **Three Sections**.
(ii) Section-A is compulsory, Section-B and C contains internal choice.

SECTION-A

0.5 x 10 = 5

1. Attempt **ALL** parts of the following questions:
 - (a) A distributed system is characterized as:
 - a. Having single global clock b. No global clock c. Synchronized clock d. All the mentioned [BT2, CO1, PO2]
[BT1, CO1, PO1]
 - (b) A distributed system is
 - a. Interconnection of networks b. An isolated system c. A transparent networked system d. All the mentioned [BT3, CO1, PO3]
 - (c) The capability of a system to adapt the increased service load is called
 - a. Scalability b. Tolerance c. Capacity d. None of the mentioned
 - (d) A distributed system has:
 - a. Shared memory b. Complete and coherent view c. No shared memory d. None [BT2, CO1, PO2]
 - (e) Transparency is defined as:
 - a. Showing all the underlying execution to users b. showing the architecture [BT3, CO1, PO3]
c. Concealment from the users d. All the mentioned [BT3, CO1, PO3]
 - (f) Devices embedded in appliances such as washing machine, hi-fi systems are
 - a. Mobile devices b. Remote devices c. Isolated devices d. Ubiquitous devices
 - (g) The happened before relation captures the _____ between two events. [BT3, CO1, PO3]
 - a. Causal dependency b. Physical dependency c. Communication dependency d. None
 - (h) Lamport's clock is based on _____ time [BT3, CO1, PO3]
 - a. Clock time b. Logical time c. Physical time d. None [BT3, CO2, PO4]
 - (i) In distributed systems, the system state can be modeled or represented by a
 - a. Status graph b. Wait-for-Graph c. TWF graph d. None of the mentioned [BT4, CO2, PO4]
 - (j) Phantom deadlocks mean
 - a. Communication deadlocks b. Resource deadlocks c. Existing deadlocks d. Nonexistent deadlocks
- What are the characteristics of a Distributed System? Explain briefly.

SECTION-B

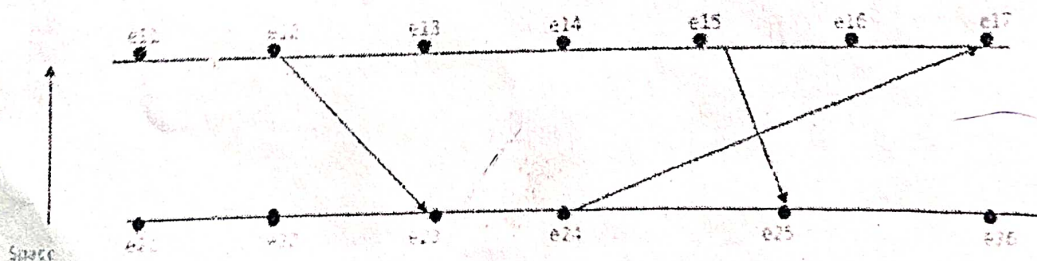
2.5 x 2 = 5

2. Attempt any **TWO** of the following questions:
 - (a) Define the inherent limitations of distributed systems. [BT1, CO1, PO2]
 - (b) Summarize the characteristics of a Distributed System? [BT2, CO1, PO3]
 - (c) Determine the role of middle ware in distributed systems [BT3, CO1, PO2]
 - (d) Illustrate the agreement protocols? Classify the agreement problems. [BT3, CO2, PO3]

SECTION-C

5 x 2 = 10

3. Attempt any **TWO** of the following questions:
 - (a) Illustrate the Lamport's clock? Assign the time stamp of every event in the figure given below. [BT-4, CO 1, PO 3]



- (b) Articulate the transparency in distributed systems? Enlist the types of transparencies exist in distributed systems. [BT3, CO1, PO2]
- (c) Prove that, if the total number of processors is 3 then the Byzantine agreement cannot be reached [BT4, CO2, PO2]
- (d) Validate the Edge-Chasing algorithm with suitable example and diagram. [BT-4, CO2, PO 3]